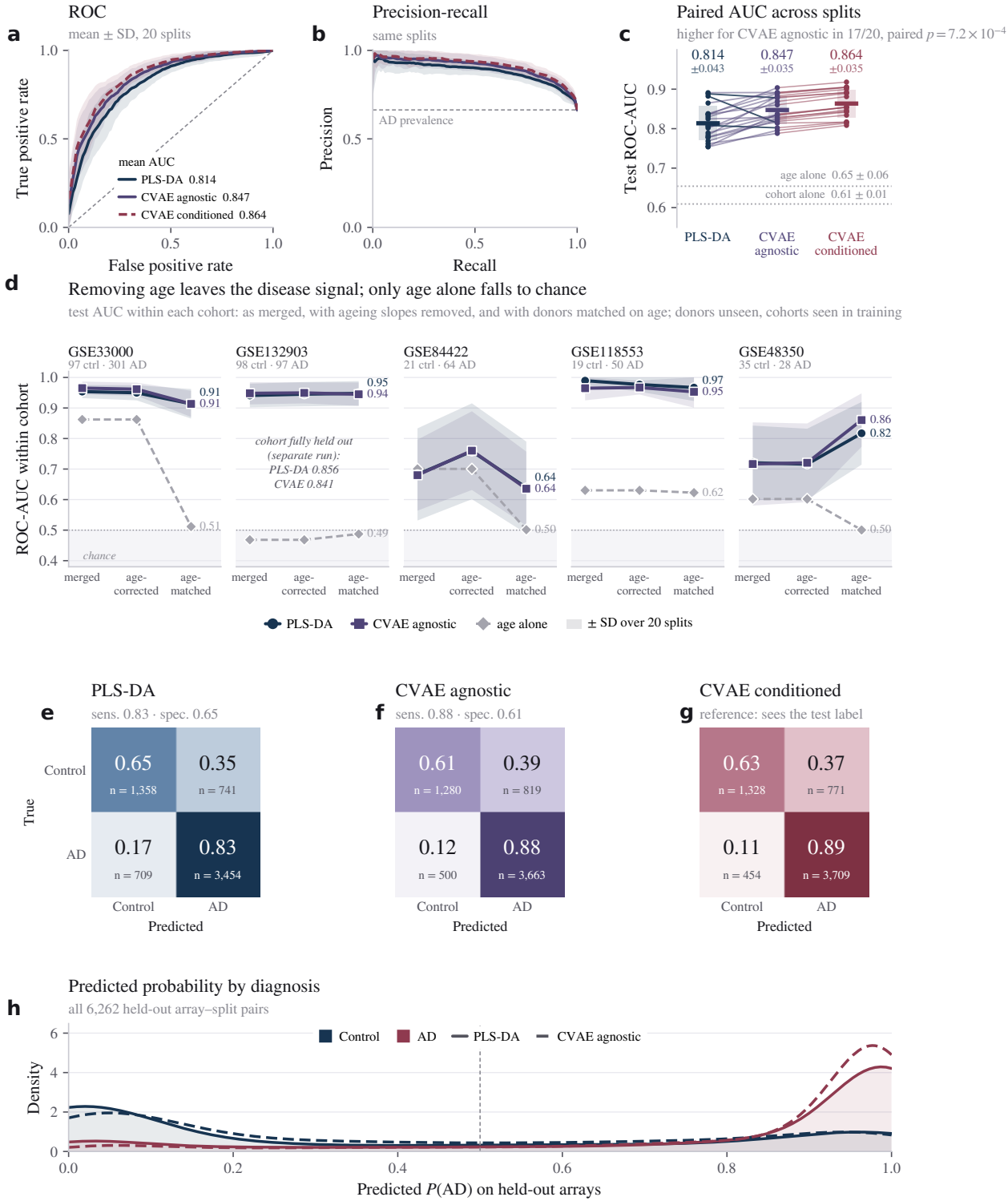

a
ROC
mean ± SD, 20 splits
True positive rate
False positive rate
mean AUC
PLS-DA 0.814
CVAE agnostic 0.847
CVAE conditioned 0.864
b
Precision-recall
same splits
Precision
Recall
AD prevalence
c
Paired AUC across splits
higher for CVAE agnostic in 17/20, paired p = 7.2 × 10−4
0.814 ±0.043
0.847 ±0.035
0.864 ±0.035
Test ROC-AUC
age alone 0.65 ± 0.06
cohort alone 0.61 ± 0.01
PLS-DA
CVAE agnostic
CVAE conditioned
d
Removing age leaves the disease signal; only age alone falls to chance
test AUC within each cohort: as merged, with ageing slopes removed, and with donors matched on age; donors unseen, cohorts seen in training
GSE33000
97 ctrl · 301 AD
GSE132903
98 ctrl · 97 AD
GSE84422
21 ctrl · 64 AD
GSE118553
19 ctrl · 50 AD
GSE48350
35 ctrl · 28 AD
ROC-AUC within cohort
cohort fully held out (separate run): PLS-DA 0.856 CVAE 0.841
chance
merged
age-corrected
age-matched
PLS-DA
CVAE agnostic
age alone
± SD over 20 splits
e
PLS-DA
sens. 0.83 · spec. 0.65
f
CVAE agnostic
sens. 0.88 · spec. 0.61
g
CVAE conditioned
reference: sees the test label
True
Control
AD
Predicted
0.65 n = 1,358
0.35 n = 741
0.17 n = 709
0.83 n = 3,454
0.61 n = 1,280
0.39 n = 819
0.12 n = 500
0.88 n = 3,663
0.63 n = 1,328
0.37 n = 771
0.11 n = 454
0.89 n = 3,709
h
Predicted probability by diagnosis
all 6,262 held-out array–split pairs
Density
Predicted P(AD) on held-out arrays